

Linear Dependence, Independence, and Spanning Sets

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Abstract

This project examines how linear dependence, linear independence, and spanning sets behave when one finite subset of a vector space is contained within another. We will define linear dependence, spanning sets, vector spaces, and more before proving two nice results.

1 Introduction

Vector spaces, spanning sets, and linear independence are fundamental concepts of Linear Algebra. There are several excellent books that expand on these topics in great detail, such as *Linear Algebra* [1]. In this book there are many great exercises, and we will prove a couple that appear on page 39. In Section 2, we define all that is needed to prove the following statements (which we will then prove in Section 3):

Let V be a vector space over a field $\mathbb{F}$. If $S_1 \subseteq S_2 \subseteq V$, then

1. If S_1 is linearly dependent, then S_2 is linearly dependent. If S_2 is linearly independent, then S_1 is linearly independent.
2. If S_1 spans V , then S_2 also spans V .

2 Definitions

Definition 2.1. A vector space V over a field $\mathbb{F}$ consists of a set on which two operations (called addition and scalar multiplication respectively) are defined so that for each pair of elements x, y , in V , there is a unique element $x + y$ in V , and for each element a in $\mathbb{F}$ and each element x in V there is a unique element ax in V , such that the following conditions hold:

- (VS1) For all x, y in V , $x + y = y + x$ (commutativity of addition).
- (VS2) For all x, y, z in V , $(x + y) + z = x + (y + z)$ (associativity of addition).
- (VS3) There exists an element in V denoted by 0 such that $x + 0 = x$ for each x in V (additive identity).
- (VS4) For each element x in V there exists an element y in V such that $x + y = 0$ (additive inverse).
- (VS5) For each element x in V , $1x = x$ (scalar identity).
- (VS6) For each pair of elements a, b in $\mathbb{F}$ and each element x in V , $(ab)x = a(bx)$ (associativity of scalars).
- (VS7) For each element a in $\mathbb{F}$ and each pair of elements x, y in V , $a(x + y) = ax + ay$ (first distributive property).
- (VS8) For each pair of elements a, b in $\mathbb{F}$ and each element x in V , $(a + b)x = ax + bx$ (second distributive property).

(This definition is taken from page 7 of Linear Algebra [1]).

Definition 2.2. Let V be a vector space over a field $\mathbb{F}$, and $S \subseteq V$. A vector $v \in V$ is a linear combination of elements of S if there exist vectors $v_1, \dots, v_k \in S$ and scalars $a_1, \dots, a_k \in \mathbb{F}$ such that

$$v = a_1v_1 + \dots + a_kv_k.$$

Definition 2.3. A subset $S \subseteq V$ is said to be spanning for V if every vector $v \in V$ can be written as a linear combination of elements of S . Equivalently written as $\text{span}(S) = V$.

Definition 2.4. A finite set $\{v_1, \dots, v_k\} \subseteq V$ is linearly dependent if there exist scalars $a_1, \dots, a_k \in \mathbb{F}$, not all zero, such that

$$a_1v_1 + \dots + a_kv_k = 0.$$

If the only case is the trivial one ($a_i = 0$ for all i), the set is linearly independent.

(This definition is explained further in Chapter 2.1 of Handbook of Linear Algebra [2])

Lemma 2.5. In a vector space V , the following statement is true: $0v = 0$ for all $v \in V$.

Proof. We know that in the field $\mathbb{F}$ for vector space V , we have that $0 + 0 = 0$. So using VS8,

$$0v = (0 + 0)v = 0v + 0v.$$

Adding the additive inverse of $0v$ to both sides (VS4), call it $-0v$, we get,

$$0v + (-0v) = 0v + 0v + (-0v) \implies 0 = 0v + 0 \implies 0 = 0v.$$

Thus, $0v = 0$. □

(There is a similar proof of this Lemma on page 12 of Linear Algebra [1])

3 Results

Theorem 3.1. Let V be a vector space over a field $\mathbb{F}$ and let $S_1 \subseteq S_2 \subseteq V$ be finite sets.

1. If S_1 is linearly dependent, then S_2 is linearly dependent. If S_2 is linearly independent, then S_1 is linearly independent.
2. If S_1 is spanning for V , then S_2 is spanning for V .

Proof. 1a) Suppose that S_1 is linearly dependent. Then there exist vectors $v_1, \dots, v_k \in S_1$ and scalars $a_1, \dots, a_k \in \mathbb{F}$, not all zero, such that

$$a_1v_1 + \dots + a_kv_k = 0.$$

Since $S_1 \subseteq S_2$, a linear combination from S_2 is of the form

$$a_1v_1 + \dots + a_kv_k + a_{k+1}v_{k+1} + \dots + a_lv_l$$

with v_i , for $i > k$, belonging to S_2 but not S_1 , and $a_i \in \mathbb{F}$. Setting this sum equal to zero, we can see that there exist vectors $v_1, \dots, v_l \in S_2$ and scalars $a_1, \dots, a_l \in \mathbb{F}$, not all zero, such that

$$a_1v_1 + \dots + a_lv_l = 0.$$

This can be achieved by choosing $a_{k+1} = a_{k+2} = \dots = a_l = 0$, so using Lemma 2.5,

$$a_{k+1}v_{k+1} + \dots + a_lv_l = 0 + \dots + 0 = 0.$$

Then $a_1v_1 + \dots + a_kv_k = 0$, with $a_1, \dots, a_k \in \mathbb{F}$ as before (not all zero as S_1 is linearly dependent). Thus, S_2 is linearly dependent.

1b) Suppose S_2 is linearly independent, and for a contradiction, S_1 is linearly dependent. By the above argument, S_2 would be linearly dependent (as $S_1 \subseteq S_2$). This contradicts the linear independence of S_2 previously stated, so S_1 must be linearly independent.

2) Suppose $S_1 = \{v_1, \dots, v_k\}$ is a spanning set for the vector space V , over field $\mathbb{F}$. This means that any $v \in V$ can be written as a linear combination of vectors from S_1 ,

$$v = a_1v_1 + \dots + a_kv_k, \quad a_i \in \mathbb{F}.$$

As $S_1 \subseteq S_2$, all the vectors in S_1 are also in S_2 . So *generally*, $S_2 = \{v_1, \dots, v_k, v_{k+1}, \dots, v_l\}$. Therefore, v can be written as

$$v = a_1v_1 + \dots + a_kv_k + a_{k+1}v_{k+1} + \dots + a_lv_l, \quad a_i \in \mathbb{F}.$$

with a_i the same as before for $i \leq k$, and $a_i = 0$ for $i > k$. Thus, any $v \in V$ can be written as a linear combination of vectors from S_2 , so S_2 is spanning for V .

□

References

- [1] S. Friedberg, A. Insel and E. Spence, *Linear Algebra (4.Ed)*, Pearson 2015.
- [2] L. Hogben, *Handbook of Linear Algebra (2.Ed)*, CRC Press 2014.